

CHETAN CHIRAG KH

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SUMMARY

Data Engineer with 3+ years of experience designing scalable ETL and ELT pipelines, dimensional data models, and enterprise BI solutions across Snowflake, Redshift, Azure, and AWS. Experienced in building data warehouses, automating workflows with Airflow and dbt, and developing Power BI and Tableau dashboards that enable data-driven decision making. Hands-on expertise in SQL, Python, cloud data platforms, and data governance, with a strong focus on delivering reliable, high-quality data products for business stakeholders.

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, T-SQL, PySpark.
- BI and Visualization:** Power BI (DAX, Power Query/M, RLS, KPI Dashboards), Excel (Advanced), Tableau, Looker, QlikView.
- ETL and Pipelines:** ETL/ELT Pipelines, Apache Airflow, dbt, Azure Data Factory, SSIS, Apache Spark, Data Integration, Pipeline Automation.
- Data Modeling and Warehousing:** Star Schema, Snowflake Schema, Dimensional Modeling, Semantic Layer, Data Warehousing.
- Cloud and Data Platforms:** Microsoft Azure (Azure Synapse, Azure Data Factory, Microsoft Fabric, OneLake), AWS (S3, EMR, Redshift), Snowflake, BigQuery.
- Databases:** SQL Server, PostgreSQL, Oracle, MySQL.
- Analytics and BI:** SQL Optimization, Data Quality, Data Governance, KPI Design, A/B Testing, Predictive Modeling, Stakeholder Reporting, Requirements Gathering.
- DevOps and Tools:** Git, Azure DevOps, CI/CD for BI, Alteryx, JIRA, Agile, Scrum, SPSS, SAS.

WORK EXPERIENCE

Business Intelligence Engineer, Walmart Global Tech, United States Aug 2025 - Apr 2026

- Designed and optimized **SQL, Python**, and **T-SQL** data pipelines on **Snowflake** and **Amazon Redshift**, transforming 5 to 10M+ retail records into curated datasets that reduced reporting latency by **25%**.
- Built enterprise **Power BI** (DAX, Power Query, RLS) dashboards tracking **18+ operational KPIs**, enabling executive reporting and self-service analytics for **10+ stakeholders**.
- Developed automated ETL pipelines using Python, Apache Airflow, dbt and Azure Data Factory, integrating REST APIs and enterprise systems while reducing manual data preparation by **30%**.
- Designed dimensional **Star Schema** data models and governed semantic layers to improve reporting consistency, data quality, and business-wide KPI standardization. Collaborated with product managers, engineers, and business stakeholders to gather requirements and to define data models.

Business Intelligence Engineer Intern, Straive, India Jan 2024 - Mar 2024

- Built governed workforce analytics solutions using **Workday, Tableau**, and **QlikView**, supporting HR reporting, compensation analysis, and executive dashboards.
- Developed enterprise **Snowflake** dimensional models using **dbt**, improving dashboard performance from **45 seconds** to **8 seconds** across 15 production reports. Automated enterprise ETL workflows using **Apache Airflow, SSIS**, and **SQL**, improving refresh frequency from daily to hourly while strengthening data validation and governance.

Data Engineer, Mphasis, India Jun 2022 - Jun 2023

- Engineered scalable **ETL/ELT pipelines** using Python, SQL, Apache Airflow, dbt, AWS (S3), and Azure Synapse Analytics to process IoT and operational datasets for enterprise reporting.
- Designed enterprise **Star Schema** and dimensional data warehouse models on **Snowflake** and **BigQuery**, enabling governed self-service analytics for **20+ business users**.
- Optimized SQL queries, automated reporting workflows, and improved enterprise data quality through reconciliation, validation, and governance practices, reducing monthly reporting cycles by **60%**.

Data Engineer, Maxgen Technologies, India Jan 2021 - Jun 2022

- Developed cloud-based ETL pipelines using **Python, SQL, SSIS, Azure Data Factory**, and **AWS (S3, EMR)**, integrating REST APIs into enterprise reporting platforms and built an interactive **Power BI** and **QlikView** dashboards for healthcare and finance clients, improving operational visibility and executive reporting.

PROJECTS

AutoCrash, Injury Severity Prediction - Python + LightGBM + SHAP + Optuna + SHAP Aug 2025 - May 2026

- Built an end-to-end predictive analytics pipeline using Python, SQL, and LightGBM on **549K+ NHTSA CRSS** crash records, achieving **80.76% accuracy** and **0.862 ROC-AUC** through feature engineering and hyperparameter optimization with Optuna.
- Applied SHAP explainability and data validation techniques to eliminate data leakage and improve model reliability. **Research accepted at INFORMS 2026.**

Self-Verifying Analytics Agentic AI - Python + DuckDB + LLM tool calling Feb 2026 - Apr 2026

- Built an autonomous agentic AI that runs SQL via LLM tool calls and verifies outputs against reconciliation checks to catch silent analytics failures like fan-out row inflation.

BI Assistant for Self-Serve Reporting - Python, Power BI REST API, Snowflake, LLM tool calling. Nov 2025 - Jan 2026

- Engineered a semantic layer over Snowflake mapping business terms to governed **SQL** and **DAX** logic, and built a natural language to SQL pipeline that validated parameterized queries against it before execution to enforce correctness over raw LLM output.
- Integrated the **Power BI REST API** to dynamically generate and embed the right visual by query type, tuning term mappings across 50+ ambiguous questions to lift first-try accuracy from **60%** to **90%**.

EDUCATION

M.S. Information Systems, California State University, Long Beach GPA 3.7, May 2026

Capstone: Insurance claims predictive modeling using Python, R, and SQL with A/B test design.

B.E. Computer Science and Engineering, Don Bosco Institute of Technology GPA 3.5, July 2022

CERTIFICATIONS and RECOGNITION

Oracle Cloud Data Science and Gen AI Professional, ChatGPT and Perplexity Fall 2025 Ambassador, SoCal Datathon Runner-Up 2024 (Alteryx, OC Transit), TA at CSULB, guided 100+ students in SQL, Python, AWS.